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Design and control of the rapid legged platform GAZELLE

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ABSTRACT

This study presents the design and system integration of the 13-degree-of-freedom legged platform, GAZELLE. The goal of this research is to develop a fast and reliable biped platform for walking experiments. Rapid leg movement can increase robustness in walking stability. During external push or walking on uneven ground, fast leg movement can realize an abrupt change on the landing position. GAZELLE is characterized by a lightweight design, a link-driven structure for low leg inertia, a wide range of motion, and a direct-stacked fin air-cooling module. The actuator is designed using a special type of harmonic drive (lightweight CSF type) and a 200 W brushless DC motor with high power density. The knee and ankle joints are link-driven; hence, the actuators can be placed on the upper position, and the leg inertia is reduced. A novel air cooler design for the cylinder-type motor, which is extremely light and highly efficient, is introduced herein and realized by stacking cooling fins to the motor housing. This study also includes a brief introduction of a stabilizing controller and the walking experiment results of GAZELLE.

1. Introduction

Man-made devises are often motivated by the design and mechanism of men themselves. A humanoid robot is a clear example of this. To improve performance and realize more adaptability in a human working environment, robots' configurations have been made very similar to humans. In the DARPA Robotics Challenge finals in 2015 (DRC finals), challenging tasks have been given assuming a radiation disaster in an environment where humans work [1]. Many bipedal humanoid robots in the DRC finals showed state-of-the-art performances [2–7]. Most tools and environments are fit for humans; hence, the development of a human-like bipedal locomotion is both necessary and important as an alternative for wheel-based robots.

Bipedal platforms have been developed and researched upon from decades ago. In many universities, bipedal platforms have been developed for locomotion study. Waseda University has performed research on the development of the WABIAN series [8]. They implemented a 6-degree-of-freedom (DOF) parallel manipulator to a leg in a WL-16- and WABIAN-RIV-mounted stereo vision system. After that, the joint position-controlled humanoid, WABIAN-2, was developed, which can walk with a stretched knee [9]. Oregon State University also developed a spring-loaded inverted pendulum model for model-based walking with its robot, ATRIAS [10]. Recently the team, now working as Agility Robotics, has developed an upgraded platform, called Cassie [11]. Their series of platforms use a series elastic actuator for the torque control of joints. Technical University of Munich developed Johnnie [12] and Lola [13]. Lola consists of a 7-DOF leg with shock-absorbing feet. It reduces its weight through optimal design and uses links to actuate the joints. KAIST began developing the KHR series [14,15], and is now working on the HUBO series [5,16].

Industries were also developing unique humanoid robots. Honda developed its ASIMO [17]. Toyota worked on a partner robot [18]. Samsung developed Roboray [19]. All these robots have shown impressive performances. Research facilities also took part in the humanoid development. AIST has maintained and updated its HRP series humanoid robot [20–22]. KIST developed MAHRU-R [23]. DLR developed the torque-controlled robot, TORO [24]. IIT developed the compliant humanoid, COMAN [25]. Different research facilities have also used humanoid platforms to develop stabilizing control, path planning, and motion generation.

Research with regard to the performance of humanoid parts, such as motor cooling, has been performed. Urata et al. [26] developed a liquid-cooling module for motor heat dissipation. Paine and Sentis [27] analyzed a motor's thermal dynamics to assess the liquid cooling efficiency. Jung et al. [5] suggested a robot design for motor heat dissipation using body frame contact and cooling fin.

The current study presents the development of a lightweight and fast biped platform. We overview the basic specifications of GAZELLE

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and briefly describe its electronic, software, and sensory systems. The electronic and software systems are constructed based on the system of DRC–HUBO+ [5], which won the first prize on the DRC finals. It showed a stable performance during the competition; hence, the reliability of the electronic and software systems has been verified. In the mechanical design part, we present a linked design of the ankle structure, which is 2-RSU (Rotation-Spherical-Universal) parallel manipulator. The kinematics and Jacobian of the 2-RSU manipulator are described, and the torque control of the ankle joint is realized using a virtual work principle. We suggest a novel method of the heat dissipater design for a cylinder-shaped motor to maximize the motor performance. This air-cooling unit performs as well as the water-cooling unit, and is extremely light and energy efficient. We conduct a walking experiment to verify the performance of the overall platform.

The rest of this study is organized as follows: Section 2 presents a brief explanation of the actuator and introduces the electronic, software, and sensory systems; Section 3 describes the special characteristics of our mechanical design; Section 4 describes the walking algorithm of GAZELLE and verifies the performance of the overall platform through the experiments; and finally, Section 5 concludes the paper.

2. System overview

2.1. Bipedal platform: GAZELLE

GAZELLE is a 13-DOF bipedal platform that is able to move fast and robustly actuated. It is designed for a bipedal walking experiment in various environments, such as uneven terrain and external disturbances (Fig. 1). Table 1 shows its overall specification. Fig. 2 presents its kinematic structure and dimensions. Although GAZELLE has a parallel manipulator structure on the ankle, it can be considered coinciding ankle pitch and roll axes.

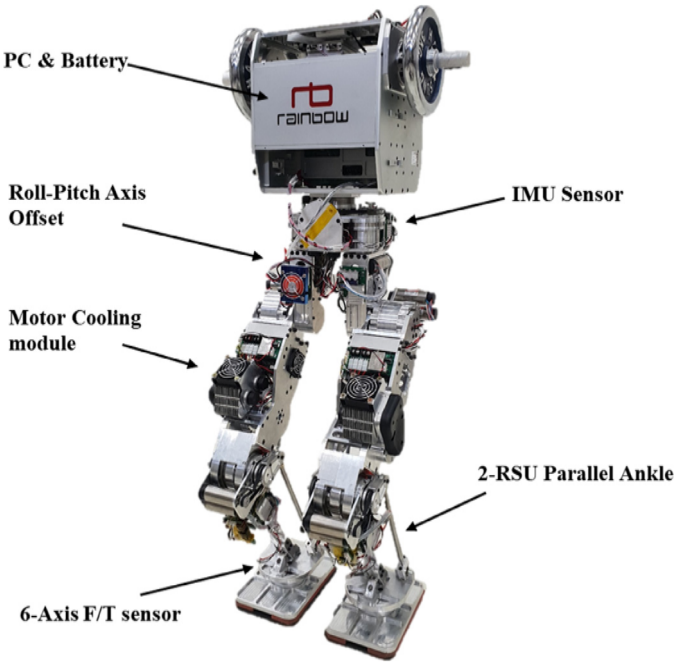


Fig. 1. Bipedal platform, GAZELLE.

2.2. Joint actuator

The actuator system consists of a motor and a harmonic drive. The CSF type of the high-capacity harmonic drive makes accurate position control with high joint stiffness possible. All motors used in the GAZELLE

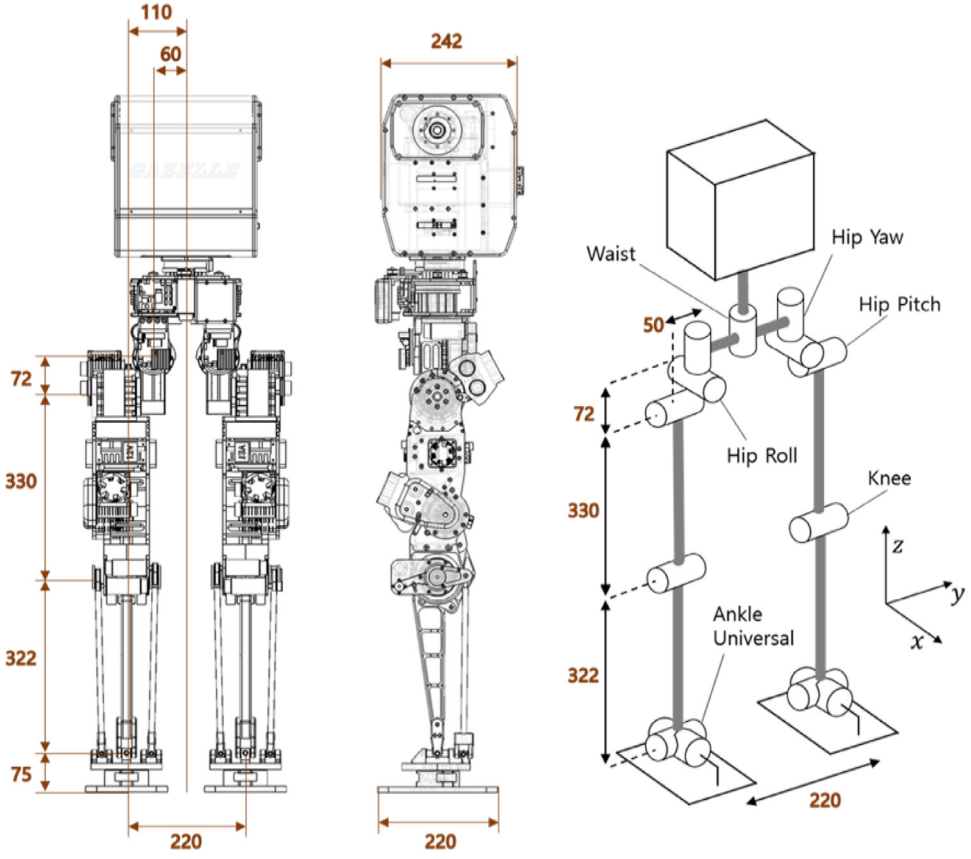


Fig. 2. Kinematic structure and dimension of GAZELLE. The ankle joint is a parallely actuated universal joint. However, we consider it as serially driven ankle pitch and roll axes, which are coinciding. This assumption will be used to solve ankle kinematics.

Table 1
Specification of GAZELLE.

Height		130 cm
Weight	Inc. battery	37 kg
	Exc. battery	33 kg
DOF	Total	13
	Leg	2 legs × 2 DOF
	Waist	1 DOF
Sensors		IMU (gyro and accelerometer), F/T sensor
Max walking speed		1.8 km/h

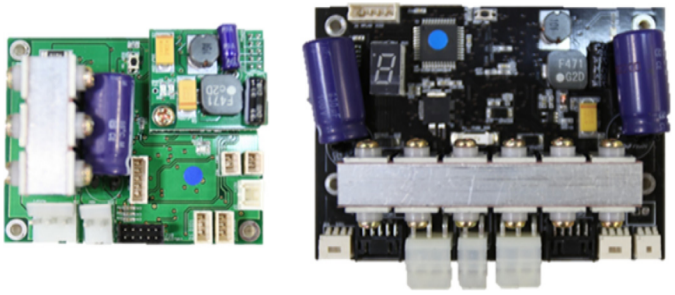


Fig. 3. 1-Channel motor drive (left) and 2-channel motor drive (right). The 2-channel motor driver can drive two motors separately, and can drive two motors as one 1-channel (dual mode). The dual mode is used for the hip pitch joint and the knee joint. In the dual-mode drive, the PWM of the two motors is shared.

system have the same specification. The model is 48 V MAXON EC 4-pole with a nominal output of 200 W. The knee joint and the hip pitch joint are driven by a double motor with an air-cooling system. The details of the air-cooling system will be described in the corresponding section. The motor position is encoded by a magnetic encoder with 1000 pulses per revolution. The absolute position of the link is initialized by a contactless proximity sensor. Table 2 shows the specifications of each joint.

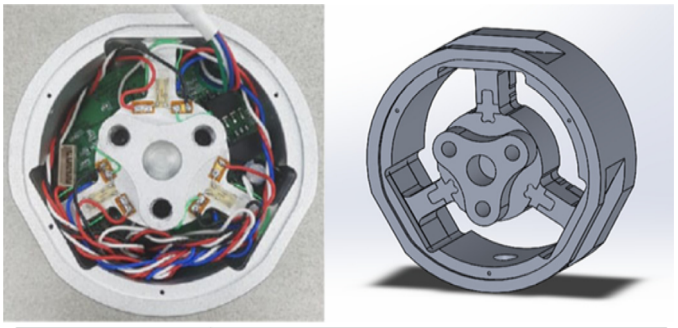
2.3. Motor driver

Our research group previously developed DRC + HUBO+ [5], which won the DRC finals. The DRC HUBO+ has stable electronic systems with stable communication. The GAZELLE mostly implements electronic systems from DRC–HUBO+. The motor driver is a specialized 48 V system that offers multiple-control mode, such as pulse width modulation (PWM), velocity, position, and current. The continuous phase current is available up to 20 A depending on the thermal condition. This driver supports CAN communication up to 1000 Hz (Fig. 3).



Gyro	Range	±450°/s
	Stability	0.8°/h in run bias
Acc. meter	Range	±8 g
	Stability	3.2μg In run bias
2KHz, CAN/RS232		

Fig. 4. Specification of the IMU sensor.



Measurement Range	Fx, Fy, Fz	±2000N
	Mx, My, Mz	±80Nm
CAN communication		

Fig. 5. Specification of the force/torque sensor.

2.4. Sensor system

2.4.1. IMU sensor

A six-axis inertial measurement system is constructed with the ADIS-16,489 model (Analog Devices) (Fig. 4). This module has a three-axis accelerometer, a three-axis gyro, and a temperature sensor. This module only supports SPI communication; hence, we developed a data acquisition (DAQ) board with PIC32MZ2048EFH MCU and apply the necessary algorithm for data fusing and communications with our software system. Two different sensors are unified by an extended Kalman filter with a varying gain. The measured orientation from the IMU is used to estimate the CoM state.

2.4.2. F/T sensor

GAZELLE uses the F/T sensor, which was specially designed for a humanoid by our research group (Fig. 5). The sensor has the following

Table 2
Joint specifications.

Joint		Gear ratio *N*			Max torque	Max speed	Angle range[a]
		Pulley	HD	total	(Nm)	(rad/s)	(deg)
Waist	Yaw	2	100	200	71	8.37	± 90
	Yaw	1	100	100	71	16.74	± 45
Hip	Roll	2	100	200	147	8.37	−45–30
	Pitch	1.7	100	170	284	9.85	−30–90
Knee	Pitch	1.2	100	120	284	13.95	0–135
Ankle[b]	Motor1	1.7	50	85	70	19.70	–
	Motor2	1.7	50	85	70	19.70	–
	Roll	–	–	55–65	140	25.7–35.4	± 45
	Pitch	–	–	75–85	140	19.7–22.3	−65–30

[a] The angle range is described based on the right leg.
[b] The ankle is a 2RSU-parallel joint. The specifications of each actuator and equivalent roll and pitch joint specifications are shown together.

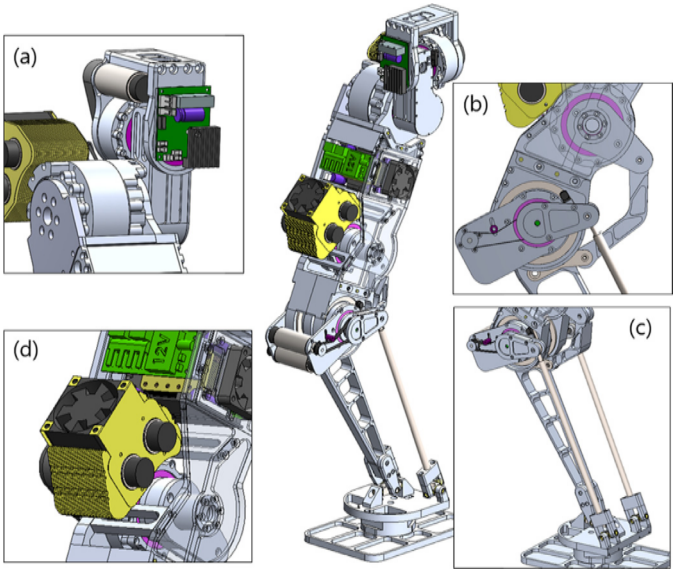


Fig. 6. Feature of the GAZELLE leg design. Hip roll and pitch joint offset (a), linked knee joint (b), 2-RSU parallel linked ankle joint (c), and direct-stacked fin air-cooling system (d).

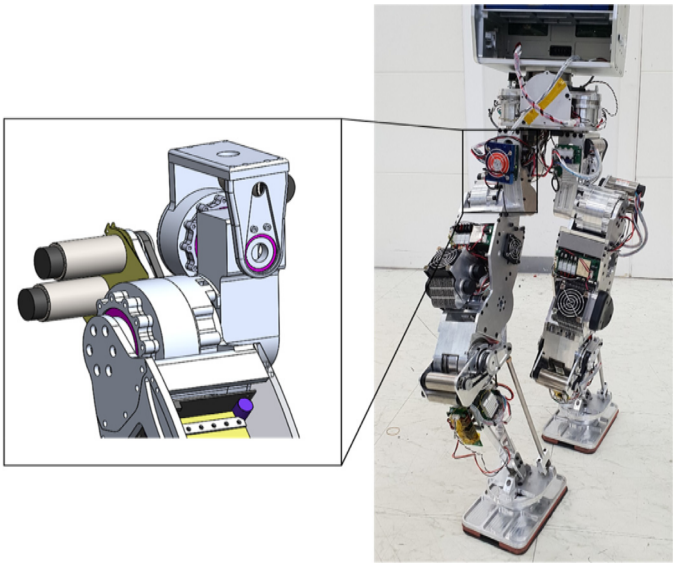


Fig. 7. Design of hip pitch and roll joint. Hip pitch and roll joint are not coinciding, so wide range of joint motion can be realized and current angle of pitch axis does not affect to roll axis angle limitation (and vice versa). Wide range of pitch and roll axis can make the configuration of one-line walking.

structure: a three-beam Maltese cross type and a measurement range of ± 2000 N (F_x, F_y, F_z) and ± 80 Nm (M_x, M_y, M_z). The semiconductor strain gauge (Micron Instruments) is applied to improve the signal-to-noise ratio. The gauge factor is 140, which is 60 times greater than that of a foil gauge. A DAQ board with seven analog–digital converter circuits with an amplifier, a three-axis accelerometer, and an MCU is embedded. The DAQ board can perform force/torque decoupling operation, and CAN communication is available. The F/T sensor is placed on the ankle and measures the ground reaction force and foot orientations.

3. Details of the GAZELLE leg

3.1. Hip and knee joints

The hip roll and pitch axes were separated to ensure the hip joint's angle range (Fig. 6(a)). By separating the hip roll axis and the pitch axis, both joints can simultaneously have a wide joint range. This leg configuration still has an analytic inverse kinematics solution because it has three successive parallel axes (i.e., hip pitch, knee, and ankle pitch) [28]. Thanks to the wide range of joint angles, we can have a motion that requires the flexibility of a hip joint, such as walking in one line (Fig. 7).

The position of the actual actuator and the point of action for the knee joint are separated by driving the four-bar linkage (Fig. 6(b)). The motor and the harmonic drive modules are located at the thigh and actuate the knee joint through the link. A bent-shaped link with sufficient stiffness is used to avoid mechanical interference. The ankle actuator can be placed on the knee position by shifting the knee actuator to the thigh position. Therefore, the 2-RSU parallel manipulator is used to actuate the ankle that will be introduced in the next section (Fig. 6(c)). The inertia of the leg with respect to the hip joint is reduced by approximately 30% by shifting up the knee and ankle actuators, with only a 5% mass increase. Reducing the inertia of the legs increases the similarity between the actual robot and the simplified models and improves the dynamic performance of the robot.

3.2. Ankle joint

The ankle joint is a 2-RSU-type parallel manipulator (Fig. 6(c)). The joint actuators of the parallel manipulator do not have to be collocated at the joints. Two motor sets exert ankle torque in parallel, increasing the maximum torque output compared to the serial configuration and greatly increasing the torque-to-weight ratio.

3.2.1. Inverse kinematics

A kinematics equation must be obtained to manipulate an object in space, in this case, the foot's orientation. This allows the end-effector position and orientation to be expressed as a function of the joint variables. [9] Gazelle's foot is controlled with a parallel manipulator. Two motors located at the knee are linked to the base of the foot with universal joints at both ends. This manipulator is classified as 2-DOF RSU parallel manipulator. Typically, the number of actuators is equal to the number of the degrees of freedom. Two motors control the pitch and roll of the foot. This section describes the inverse kinematics of the 2-RSU parallel manipulator. In Fig. 8, the angle of the drive arm A_iB_i (θ_1 and θ_2) will be expressed with the roll and pitch angles of the ankle joint (ψ and ϕ, respectively).

From the base origin {O} located at the knee joint to the rotating platform {F}, the rotation matrix is as follows:

$$ {}^{O}\mathbf{R}_{F} = \begin{bmatrix} u_x & v_x & w_x \\ u_y & v_y & w_y \\ u_z & v_z & w_z \end{bmatrix} \tag{1} $$

However, we know that the platform will rotate with only the roll and pitch angles. From the assumption that the ankle joints are serially driven in the ankle pitch and roll order, the rotational matrix using Euler angle representation with roll and pitch angles is as follows, where ψ is the roll angle, and ϕ is the pitch angle:

$$ {}^{O}\mathbf{R}_{F} = \begin{bmatrix} c\phi & s\phi c\psi & s\phi c\psi \\ 0 & c\psi & -s\psi \\ -s\phi & c\phi s\psi & c\phi s\psi \end{bmatrix} \tag{2} $$

The position vector $\overrightarrow{{}^{O}OB_i}$ from the origin of the base frame {O} to the end of the revolute joint with respect to the base frame is presented as follows:

$$ \overrightarrow{{}^{O}OB_1} = \left[-l_1 c\theta_1, \quad w_1, \quad l_1 s\theta_1\right]^{\mathrm{T}} \tag{3} $$

$$ \overrightarrow{{}^{O}OB_2} = \left[-l_2 c\theta_2, \quad -w_1, \quad l_2 s\theta_2\right]^{\mathrm{T}} \tag{4} $$

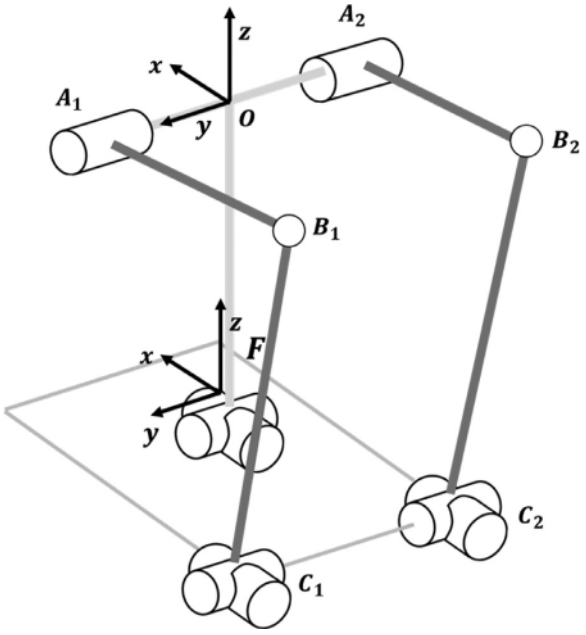


Fig. 8. Schematic drawing of the 2-RSU parallel ankle joint. The input angle is θ_1 and θ_2 at $A_1$and A_2, respectively. The output angle is ψ(roll) and ϕ (pitch) at F.

where w_i is the distance from O to A_i, and l_1 is the length between A_i and B_i. The position vectors $^F\overrightarrow{FC_i}$ in the foot frame $\{F\}$ can be presented as:

$$^F\overrightarrow{FC_1} = \left[-C_{1,x}, \quad C_{1,y}, \quad C_{1,z}\right]^{\mathrm{T}} \tag{5}$$

$$^F\overrightarrow{FC_2} = \left[-C_{2,x}, \quad -C_{2,y}, \quad C_{2z}\right]^{\mathrm{T}} \tag{6}$$

The length of link l_2, which is the distance between B_i and C_i ($i = 1,\ 2$), is given as:

$$\overrightarrow{B_iC_i} = \overrightarrow{OF} + \overrightarrow{FC_i} - \overrightarrow{OB_i} \tag{7}$$

$$l_2{}^2 = \left\|\overrightarrow{B_iC_i}\right\|^2 = \left\| l_3 + {}^O\mathbf{R}_F\,{}^F\overrightarrow{FC_i} - {}^O\overrightarrow{OB_i}\right\|^2 \tag{8}$$

where l_3 is the distance between $\{O\}$ and $\{F\}$. Two decoupled equations of θ_1 and θ_2 are derived by solving (8).

$$0 = \mathbf{A}_i + \mathbf{B}_i c\theta_i + \mathbf{C}_i s\theta_i \quad (i = 1,\ 2) \tag{9}$$

where

$$\begin{aligned}
\mathbf{A}_1 &= c_x^2 + c_y^2 + c_z^2 + l_1^2 - l_2^2 + l_3^2 + w_1^2 - 2c_y w c\psi \\
&\quad + 2c_z w_1 s\psi - 2c_z l_3 c\psi c\phi - 2c_y l_3 s\psi c\phi - 2c_x l_3 s\phi \\
\mathbf{A}_2 &= c_x^2 + c_y^2 + c_z^2 + l_1^2 - l_2^2 + l_3^2 + w_1^2 - 2c_y w c\psi \\
&\quad - 2c_z w_1 s\psi - 2c_z l_3 c\psi c\phi + 2c_y l_3 s\psi c\phi - 2c_x l_3 s\phi \\
\mathbf{B}_1 &= -2c_x l_1 c\phi + 2c_z l_1 c\psi s\phi + 2c_y l_1 s\psi s\phi \\
\mathbf{B}_2 &= -2c_x l_1 c\phi + 2c_z l_1 c\psi s\phi - 2c_y l_1 s\psi s\phi \\
\mathbf{C}_1 &= 2l_1 l_3 - 2c_x l_1 s\phi - 2c_z l_1 c\psi c\phi - 2c_y l_1 s\psi c\phi \\
\mathbf{C}_2 &= 2l_1 l_3 - 2c_x l_1 s\phi - 2c_z l_1 c\psi c\phi + 2c_y l_1 s\psi c\phi
\end{aligned} \tag{10}$$

We then introduce a new variable, t_i, and substitute this to the equation:

$$t_i = \tan\frac{\theta_i}{2} \quad c\theta_i = \frac{1 - t_i^2}{1 + t_i^2} \quad s\theta_i = \frac{2t_i}{1 + t_i^2} \tag{11}$$

$$(\mathbf{A}_i - \mathbf{B}_i)t_i^2 + 2\mathbf{C}_i t_i + \mathbf{A}_i + \mathbf{B}_i = 0 \tag{12}$$

From here, we can use a quadratic formula to solve for t_i. Once t_i is found, θ_i is found from back substitution.

3.2.2. Forward kinematics

The forward kinematics of a parallel manipulator is usually much more complex than its inverse kinematics. Forward kinematics has two approaches, namely analytic and differential. Accordingly, (9) can be organized in terms of ψ and ϕ using several substitutions; however, the analytic solution of the forward kinematics of the 2-RSU parallel manipulator will lead us to the problem of the 8th-order polynomial equation. This 8th-order polynomial equation has two double real roots and six imaginary roots. We used the differential approach to solve the forward kinematics because of the complexity of the analytic approach.

The inverse of the Jacobian matrix is necessary for the differential forward kinematics because the Jacobian matrix is necessary for the differential inverse kinematics of serial manipulators. To derive the Jacobian of the ankle joint, we start from the original constraint (7) and convert it to the velocity constraint as follows:

$$\omega_{A_iB_i} \times \overrightarrow{A_iB_i} + \omega_{B_iC_i} \times \overrightarrow{B_iC_i} = \omega_{foot} \times \overrightarrow{FC_i} \tag{13}$$

Substituting the rotational velocity of known parameters, we obtain:

$$\dot\theta_i\,\hat{\mathrm{j}} \times \overrightarrow{A_iB_i} + \omega_{B_iC_i} \times \overrightarrow{B_iC_i} = (\dot\psi\hat{\mathrm{i}} + \dot\phi\,\hat{\mathrm{j}}) \times \overrightarrow{FC_i} \tag{14}$$

where $\dot\theta$ is the angular velocity of the revolute joint, and $\dot\phi$ and $\dot\psi$ are the rotational velocities of the foot's pitch and roll angles, respectively. Now, to eliminate $\omega_{B_iC_i}$ from the above equation, we dot product $\overrightarrow{B_iC_i}$ both sides of (14) and use cross product identity (i.e., $a \cdot (b \times c) = b \cdot (c \times a)$).

$$\dot\theta_i\,\hat{\mathrm{j}} \cdot (\overrightarrow{A_iB_i} \times \overrightarrow{B_iC_i}) = (\dot\psi\hat{\mathrm{i}} + \dot\phi\,\hat{\mathrm{j}}) \cdot (\overrightarrow{FC_i} \times \overrightarrow{B_iC_i}) \tag{15}$$

We obtain the following equation from (15) and by applying dot product. This relation transforms the end-effect rotational velocity ($\dot\psi$ and $\dot\phi$) to the actuator velocity ($\dot\theta_1$ and $\dot\theta_2$).

$$\begin{bmatrix} (\overrightarrow{A_1B_1} \times \overrightarrow{B_1C_1})_y \dot\theta_1 \\ (\overrightarrow{A_2B_2} \times \overrightarrow{B_2C_2})_y \dot\theta_2 \end{bmatrix} = \begin{bmatrix} (\overrightarrow{FC_1} \times \overrightarrow{B_1C_1})_x & (\overrightarrow{FC_1} \times \overrightarrow{B_1C_1})_y \\ (\overrightarrow{FC_2} \times \overrightarrow{B_2C_2})_x & (\overrightarrow{FC_2} \times \overrightarrow{B_2C_2})_y \end{bmatrix} \begin{bmatrix} \dot\psi \\ \dot\phi \end{bmatrix} \tag{16}$$

We can now express (16) as a Jacobian form of $\mathbf{J}\dot{\mathbf{q}} = \dot{\mathbf{x}}$. The inverse of the Jacobian matrix is as follows:

$$\mathbf{J}^{-1} = \begin{bmatrix} \frac{(\overrightarrow{FC_1} \times \overrightarrow{B_1C_1})_x}{(\overrightarrow{A_1B_1} \times \overrightarrow{B_1C_1})_y} & \frac{(\overrightarrow{FC_1} \times \overrightarrow{B_1C_1})_y}{(\overrightarrow{A_1B_1} \times \overrightarrow{B_1C_1})_y} \\ \frac{(\overrightarrow{FC_2} \times \overrightarrow{B_2C_2})_x}{(\overrightarrow{A_2B_2} \times \overrightarrow{B_2C_2})_y} & \frac{(\overrightarrow{FC_2} \times \overrightarrow{B_2C_2})_y}{(\overrightarrow{A_2B_2} \times \overrightarrow{B_2C_2})_y} \end{bmatrix} \tag{17}$$

where $\overrightarrow{FC_i}$, $\overrightarrow{A_iB_i}$, and $\overrightarrow{B_iC_i}$ are defined by the joint angle (θ_1 and θ_2) and the foot orientation from (3), (4), (5), (6), and (7). Now that we have the Jacobian matrix for our differential kinematics, we will devise a feedback loop for the differential kinematics (Fig. 9). Using this closed loop differential kinematics, we find the end effector orientation by effectively taking an educated guess (estimate) number of times using the Jacobian matrix.

3.2.3. Torque control of the ankle joint

The torque control of the ankle is necessary for a humanoid to adapt walking on uneven ground. We will apply herein the virtual work principle to apply the desired torque from our robot foot. The mass and the inertia of the foot link are negligible because it is light, and does not move fast. The relationship between the motor torque and the end effector torque can be described as follows by the virtual work principle:

$$\vec{\tau}_{\mathrm{motor}} = \mathbf{J}^{\mathrm{T}}\vec{\tau}_{\mathrm{end}} \tag{18}$$

where $\vec{\tau}_{\mathrm{end}} = [\tau_{\mathrm{roll}}\ \tau_{\mathrm{pitch}}]^{\mathrm{T}}$ is the roll/pitch torque of the ankle, and $\vec{\tau}_{\mathrm{motor}} = [\tau_{\mathrm{motor1}}\ \tau_{\mathrm{motor2}}]^{\mathrm{T}}$ is the motor torque. Fig. 10 shows the torque control loop of the ankle joint. The feedforward control input was obtained from (18). The F/T sensor was used in the feedback loop.

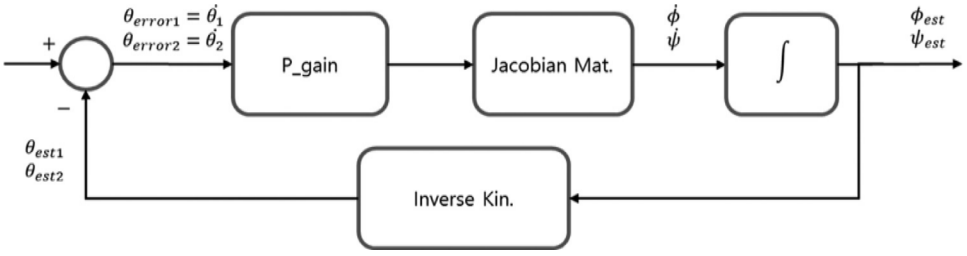


Fig. 9. Differential forward kinematics feedback loop. ϕ_{est} and ψ_{est} are the output of the differential forward kinematics loop that are the measured value of the ankle pitch and roll. θ_{est1} and θ_{est2} are the estimated motor angle from the inverse kinematics of ϕ_{est} and ψ_{est}, respectively. After several feedback processes, ϕ_{est} and ψ_{est} are converged to the real ankle pitch and roll angle.

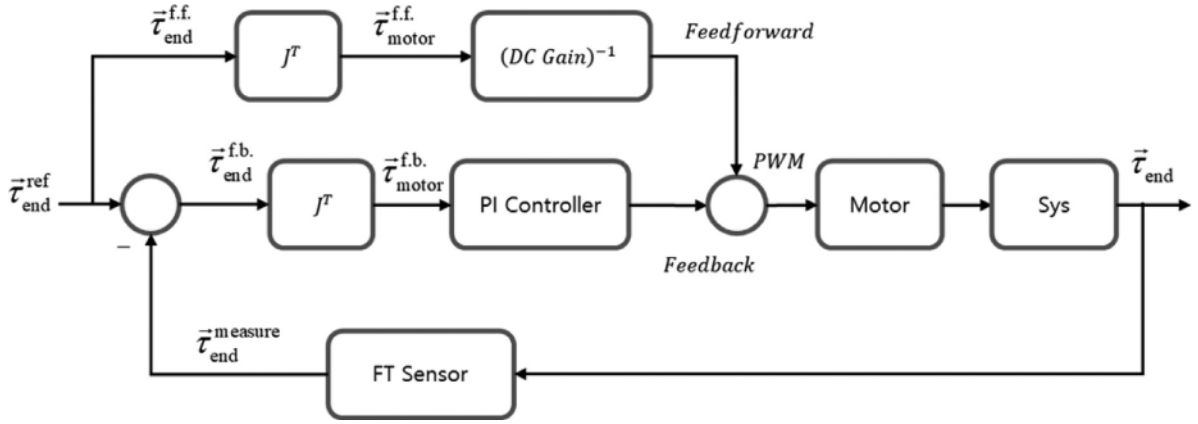


Fig. 10. Torque control loop of the ankle joint that contains feedback control loop and feedforward control loop. The feedback is dependent on the F/T sensor measurement.

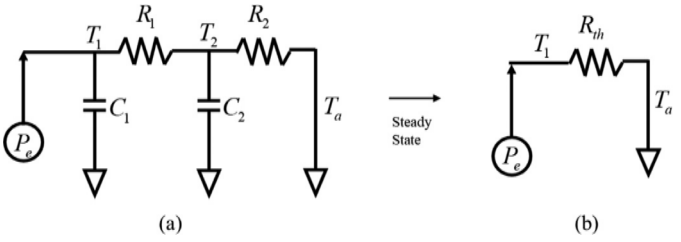


Fig. 11. Motor thermal dynamics model. P_e denotes the heat produced by the motor core. T_a is the ambient temperature. T_1 is the motor core temperature. R_1 is the motor core to housing thermal resistance. T_2 is the motor housing surface temperature. R_2 is housing to ambient thermal resistance. C_1 and C_2 are the thermal capacity of the motor core and the motor housing, respectively. The thermal circuit can be simplified for the steady state. R_{th} is the total thermal resistance of the motor to housing and housing to ambient ($R_1 + R_2$).

3.3. Heat dissipation design

More current should flow through the motor to exert more torque. However, more current through the motor generates more heat, and when too much heat is generated, the motor is burned out. Therefore, a high performance requires a more efficient heat dissipation from the motor. The novel cooling module developed herein (Fig. 6(d)) allows for a performance boost of the motor by analyzing its thermal dynamics.

3.3.1. Motor thermal dynamics model

This section briefly presents an overview of the underlying concept used in designing the heat dissipation module for the motor. A thermal dynamics model is necessary to understand how much increase in performance is achievable through better heat dissipation [27]. Fig. 11(a) shows the motor's thermal circuit. The corresponding thermal dynamics model is also derived. The heat produced from the motor core (P_e) can be expressed as:

$$\begin{aligned} P_e &= I^2 R_e(T_1) \\ R_e(T_1) &= R_0(1 + \alpha(T_1 - T_0)) \\ \alpha &\approx 0.0039\Omega/^\circ K \text{ (copper)} \\ R_0 &= 0.386 \text{ at } T_0 = 25^\circ\text{c} \end{aligned} \tag{19}$$

where R_0 is the nominal resistance at nominal temperature, and α is the material's temperature coefficient. We can derive a thermal model equation from the thermal circuit.

$$\begin{aligned} \frac{dT_1}{dt} &= \frac{1}{C_1}\left[P_e - \frac{T_1 - T_2}{R_1}\right] \\ \frac{dT_2}{dt} &= \frac{1}{C_2}\left[\frac{T_1 - T_2}{R_1} - \frac{T_2 - T_a}{R_2}\right] \end{aligned} \tag{20}$$

To simplify the problem, we can assume that the thermal circuit is in the steady state. The thermal circuit can then be simplified (Fig. 11(b)). The motor core temperature for the simplified thermal model can be calculated using the simple Ohm's law.

$$P_e = \frac{T_1 - T_2}{R_1} = \frac{T_2 - T_a}{R_2} = \frac{T_1 - T_a}{R_1 + R_2} = \frac{T_1 - T_a}{R_{th}} \tag{21}$$

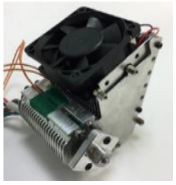
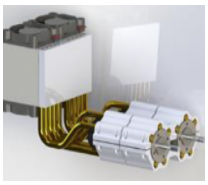
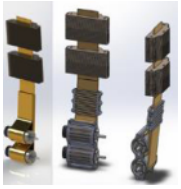
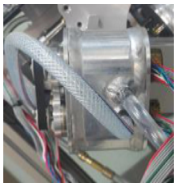
The main purpose of this section is to maximize the motor current. With the maximum operating temperature of the motor core ($T_{1\max}$) fixed and determined by the motor core material, the relationship between thermal resistance (R_{th}) and maximum allowable current can be derived from (19) and (21).

$$I_{\max} = \sqrt{\frac{T_{1\max} - T_a}{R_e(T_{1\max})R_{th}}} \quad (T_{1\max} \approx 155^\circ\text{c}) \tag{22}$$

3.3.2. Experiments for various cooling modules

Experiments were conducted to measure R_2 for various cooling module prototypes. Some of them were previously developed for DRC–HUBO+ [5]. To measure R_2 in the steady state, the current passing through the motor core (I), temperature of the motor housing (T_2), and

Table 3
Comparison of various cooling systems.

Cooling system	Motor surface	DRC-HUBO+	Heat pipe	Heat panel	Water cooling
					
Weight	0 g	380 g	485 g	225 g	700 g~
Energy	0 W	2.5 W (1-fan)	4 W (2-fan)	4 W (2-fan)	8 W (fan, pump)
R_2	7.4 K/W	0.504 K/W	0.459 K/W	0.598 K/W	0.299 K/W
R_{th}	7.61 K/W	0.714 K/W	0.669 K/W	0.808 K/W	0.509 K/W
I_{max}	5.4 A	17.7 A	18.3 A	16.6 A	21.0 A

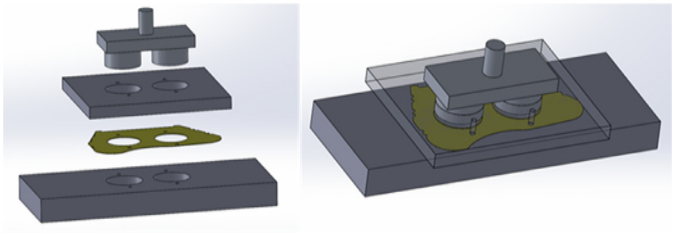


Fig. 12. Burring process of the direct-stacked fin.

ambient temperature (T_a) must be measured. When T_2 and I are given, the motor core temperature (T_1) is calculated using (19) and (21).

$$T_1 = \frac{I^2 R_0 R_1 (1 - \alpha T_0) + T_2}{(1 - I^2 R_0 R_1 \alpha)} \tag{23}$$

$R_2$is then calculated as follows with the givenT_a.

$$R_2 = \frac{T_1 - T_a}{I^2 R_e(T_1)} - R_1 \tag{24}$$

Table 3 shows the measured R_2 of various cooling systems. For comparison, we also measured R_2 of the bare surface of the MAXON motor. The cooling module for DRC–HUBO+ was based on air convection with an aluminum plate that was 1 mm thick [5]. The heat pipe and the heat panel cooling system were also based on air convection, but the heat pipe and the heat panel were used for heat transfer. Therefore, these systems are efficient when the motor mounting spot is narrow. However, the heat pipe and the heat panel should be aligned vertically to maximize the heat transfer efficiency. Although the water-cooling system showed the best performance, it also consumed the highest energy, and was the heaviest.

3.3.3. *Direct-stacked fin cooler*

A light-weight and highly efficient air-cooling module was proposed herein for a cylinder-shaped motor. In many CPU-cooling systems today, the heat pipe is usually used to transfer heat from the CPU to a stack fin, where heat is dissipated. During this process, the contacts between the CPU and the heat pipe and between the heat pipe and the stack fin cause thermal resistance. We proposed a novel air cooler design that directly stacks cooling fins to the motor surface to reduce the thermal resistance from the contacts. Holes on the fin were made through a burring process to ensure a large contact area, which reduces the thermal resistance. The burring process includes punching the hole with a punching tool bigger than the hole to make a flange around the hole. Fig. 12 shows the custom-made burring tool.

The cooling fin was made from a sheet of aluminum with a 0.3 mm thickness and burred based on the motor diameter. Burred fins were stacked on the motor one by one after applying thermal grease (Fig. 13).

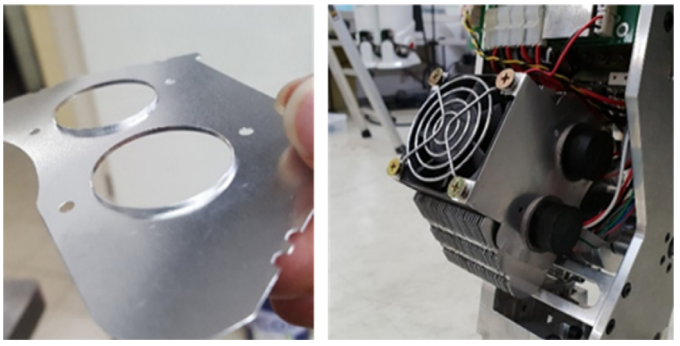


Fig. 13. Direct-stacked fin air-cooling module for the dual motor (knee joint).

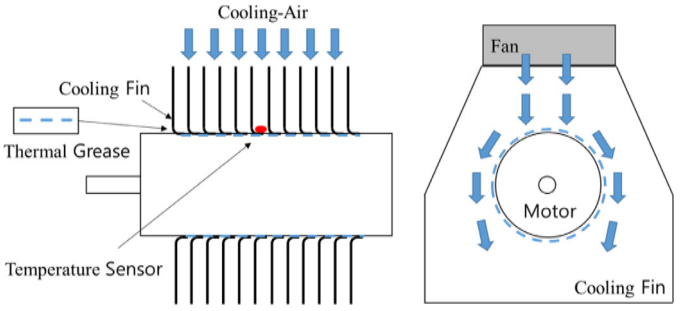


Fig. 14. Schematic drawing of the air direct-stacked fin cooling module. Approximately 35 cooling fins can be stacked for the MAXON EC 4-pole 200 W motor.

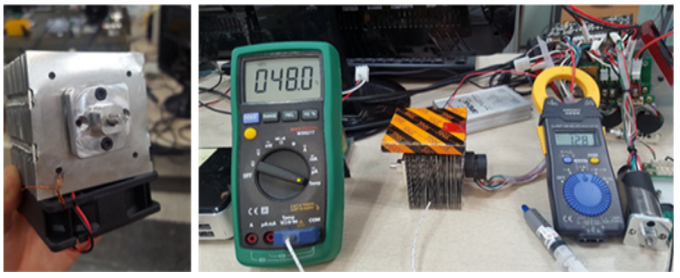


Fig. 15. Thermal resistance measurement experiment setup.

The thermal resistance between the motor housing and the ambient temperature (R_2) was measured to verify the thermal performance of the direct-stacked fin (Fig. 14). The motor current was measured with the current probe (Fig. 15). The motor housing temperature was measured with the thermal couple. Table 4 shows the experimental results.

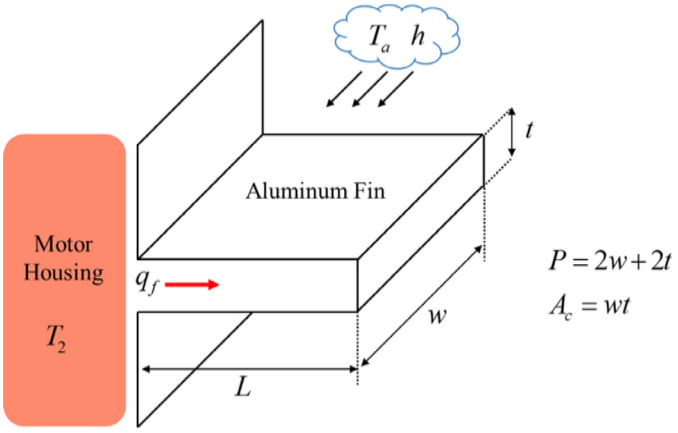


Fig. 16. Uniform cross-section fin analysis performed to find the fin length suitable for efficient heat dissipation.

The experimental results indicated that the thermal resistance (R_2) of the direct fin cooling module was approximately 0.33 K/W. The thermal performance as not as good as water cooling, but the thermal performance-to-weight ratio was much higher (i.e., weighing 130 g). It also required less power and consumed only 2.5 W for the cooling fan.

Table 4
Experimental result of the direct-stacked fin air-cooling module.

Current (A)	T_2 (°C)	R_2 (K/W)	R_{th} (K/W)	I_{max}(A)
9.0	36.3	0.338	0.548	20.22
12.8	49.6	0.336	0.546	20.21
13.4	52.0	0.332	0.542	20.30
14.0	54.7	0.330	0.540	20.34
17.7	79.5	0.335	0.545	20.25

Although it made the motor module bulky, this was the appropriate cooling module for the proposed project considering weight and energy efficiency. By adding 130 g of weight and 2.5 W of additional power, the motor can safely apply four times the normal current, denoting that it can apply four times more torque for the same RPM.

3.3.4. *Design criteria for direct-stacked fin*

The main philosophy of the direct-stacked fin is to drastically reduce the heat transfer processes of various media, thereby reducing the total thermal resistance and creating efficient and light heat-dissipation equipment. For efficient heat dissipation, the sizes and shapes of the fins must be designed properly, according to certain criteria, which are described below. The material of the fin was selected to be from the aluminum 20xx family, and its thickness was set to 0.3 mm. Aluminum

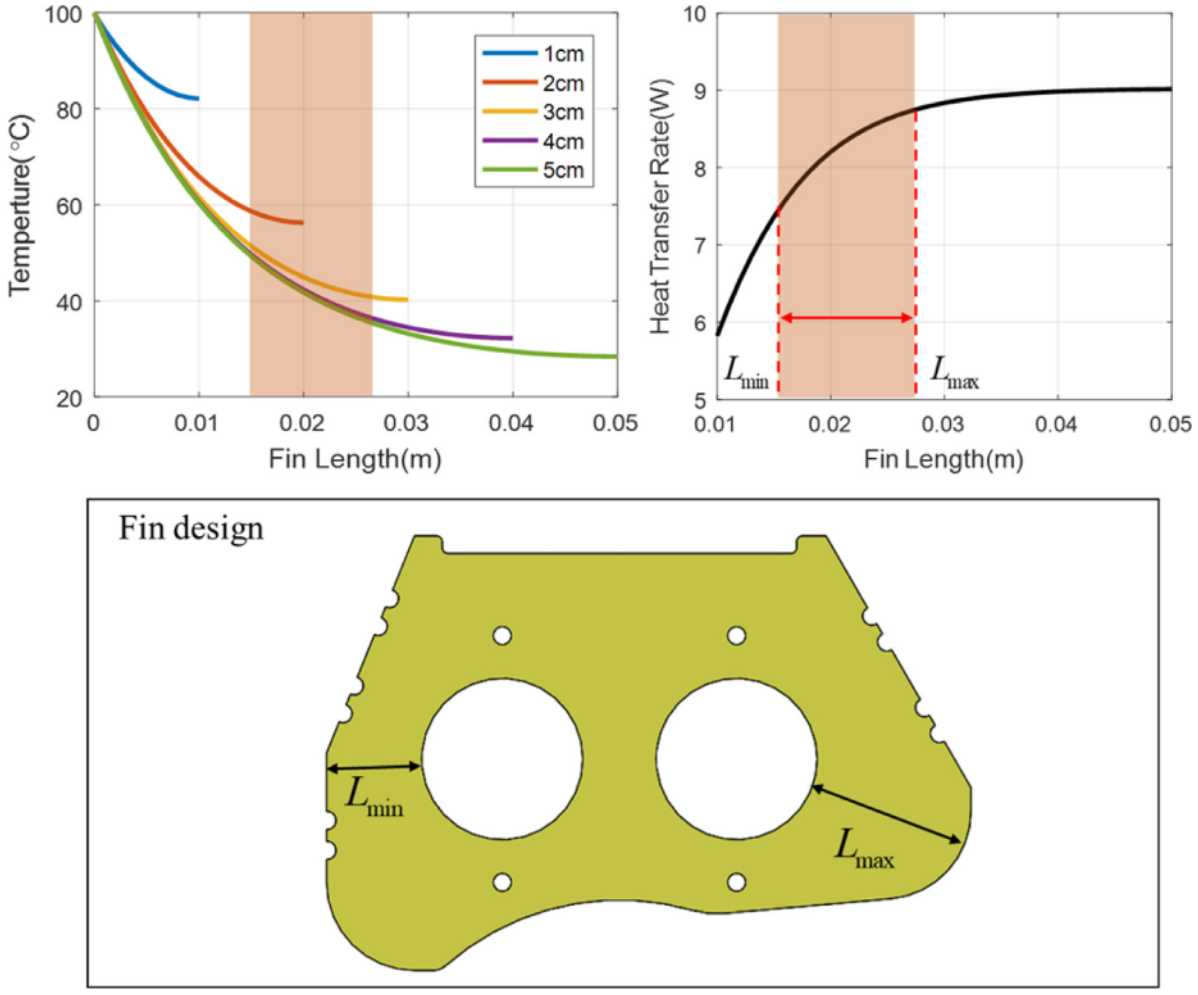


Fig. 17. Temperature distribution of uniform cross-section fin and its heat-transfer rate depending on the length of the fin (top-left). As the length of the fin increases, the temperature of the fin decreases; therefore, the heat transfer from the fin does not increase for a longer fin (top-right). We designed the fin as depicted in the figure (bottom), with length between 16 mm and 28 mm.

materials were selected considering their machinability and weight. The thickness was determined heuristically for the burring process, without causing breakage of the aluminum sheet.

The spacing between the fins is also important. If the air pressure applied by the fan is sufficient, convection occurs even if the gap between the fins is narrow. Therefore, it is important to determine the length of the fins, as the spacing can be adjusted easily when stacking the fins on the motor. As the length of the fin increases, the surface area will increase. However, as the temperature decreases toward the end of the fin, heat dissipation will not occur near end of fin. If the shape of the fin is assumed to be a simple shape, and the fin thickness, material, and convection coefficient have been determined, the most efficient length for heat dissipation can be calculated. Although the actual fin shape is cylindrical, in this work, we assume the fin to possess a uniform cross-sectional area with 30 mm width (motor diameter is 30 mm) for simplifying the analysis (Fig. 16). We assume the convection coefficient of air to be 100 $W/m^2K$and the temperature distribution and total heat transfer are calculated according to the length. (Typically, the maximum convection coefficient that can be achieved through air flow is 100 W/m^2K.)

Using boundary conditions wherein the amounts of convection heat transfer and conduction heat transfer occurring at the end of the fin are the same, the temperature distribution of the fin can be obtained as in (25).

$$\begin{aligned} \frac{\theta_{\text{fin}}(x)}{\theta_2} &= \frac{\cosh(m(L-x))+{}^{h}/_{mk}\sinh(m(L-x))}{\cosh(mL)+{}^{h}/_{mk}\sinh(mL)} \\ \theta_{\text{fin}} &= T_{\text{fin}} - T_a \\ \theta_2 &= T_2 - T_a \\ m^2 &\equiv hP/kA_c \end{aligned} \tag{25}$$

A_c is the cross-sectional area of the fin and P is the circumference of the fin. h is the convection coefficient of ambient air and k is the conduction coefficient of the fin.

$$\begin{aligned} q_{\text{fin}} &= M\frac{\sinh(mL)+{}^{h}/_{mk}\cosh(mL)}{\cosh(mL)+{}^{h}/_{mk}\sinh(mL)} \\ M &= \sqrt{hPkA_c}\theta_2 \end{aligned} \tag{26}$$

Eq. (26) gives the total heat transferred from the fin, according to the fin length L . Fig. 17 shows the temperature distribution of the fin and total heat dissipated from the fin for various fin length. We assumed the motor housing temperature to be 100°C and the ambient air temperature to be 25°C. Fig. 17 shows that the amount of heat dissipated does not increase significantly over 2 cm. Our fin is designed based on this criteria as well as spatial limitations.

3.4. Analysis of GAZELLE leg

3.4.1. Moment of inertia with respect to hip and knee joint

We used the 2-RSU parallel link structure for the ankle joint of GAZELLE. This link structure allows the motor and speed reducer to be placed on the knee instead of on the ankle. Using the link structure can increase the overall weight slightly, but the moment of inertia of the legs can be reduced. To compare the leg inertias with respect to the hip and knee joints, we used the design tool (SolidWorks) to place the ankle actuator at the ankle position and knee actuator at the knee position, temporarily (Fig. 18). The weights of both legs were approximately the same (~10 kg) but their inertias were different. The moment of inertia of the GAZELLE leg with respect to the hip joint was 1.119$kg \cdot m^2$, whereas that of the conventional leg was 1.583$kg \cdot m^2$. The inertia about the hip joint was reduced to 70.7% and the inertia about the knee joint was reduced to 54.6% (0.339$kg \cdot m^2$ to 0.185$kg \cdot m^2$) by using link structure for the ankle joint.

3.4.2. Dynamic effect of GAZELLE leg

Many bipedal robots generate walking patterns based on the linear inverted pendulum model (LIPM) [29–33]. The capture point, which is used by many walking stabilizer, is also derived from the LIPM [34, 35]. GAZELLE generates the gait pattern using an LIPM-based preview controller [32], and the controllers described in the following chapters are based on LIPM. Therefore, if the mass distribution of the robot is similar to that of LIPM, there is an advantage in securing the stability of walking. When the mass of the leg is increased or the inertia is increased, the swing leg motion will affect the walking stability. Reducing the leg inertia, as achieved in the current GAZELLE design, will fit better with an LIPM-based gait pattern and a controller that does not consider the swing leg.

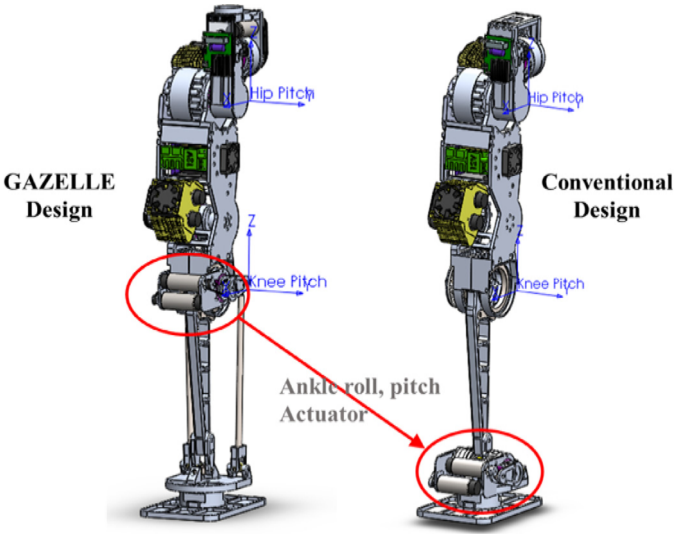


Fig. 18. Design simulation of ankle joint. We virtually moved the ankle actuator to the ankle position, in a conventional way, to compare the moments of inertia of the leg with respect to the hip and knee joints.

We simulated two scenarios using a full dynamics simulator (Choreonoid [36]), to verify the effectiveness of the GAZELLE design. The first was to walk with the model reflecting the physical properties of the designed GAZELLE platform and the second was to walk with the model reflecting the physical properties of the conventional leg with relatively high inertia, as described in Fig. 18. In both cases, we compared the measured ZMP with the planned reference ZMP. On flat ground, if the actual ZMP track the reference ZMP well, it meant that the walking is stable [37].

In order to better observe the effect of leg inertia, we let the robot to walk at relatively high speeds with 30 cm stride in a period of 0.5 s (2.16 km/h). For the first scenario, which is that of the proposed GAZELLE leg, the ZMP RMS error during the entire walking time was 0.0480 m, and for the second scenario, the ZMP RMS error was 0.0763 m (Fig. 19). The robot model with high leg moment of inertia showed 59% larger ZMP error. In the case of fast walking, the forward stepping motion had a great effect on the ZMP fluctuation in sagittal direction. Therefore, the ZMP error in the sagittal direction was larger than the ZMP error in the lateral direction (Figs. 20, 21). From this result, we can conclude that the GAZELLE leg designed in this study is suitable for rapid and stable walking.

4. Walking pattern and stabilization control

The basic components of walking are the walking patterns and the controllers that stabilize it. Walking patterns are generated based on the robot model, while controllers are constructed based on the sensor feedback. This section will briefly introduce the walking pattern generation method and the stabilization controller.

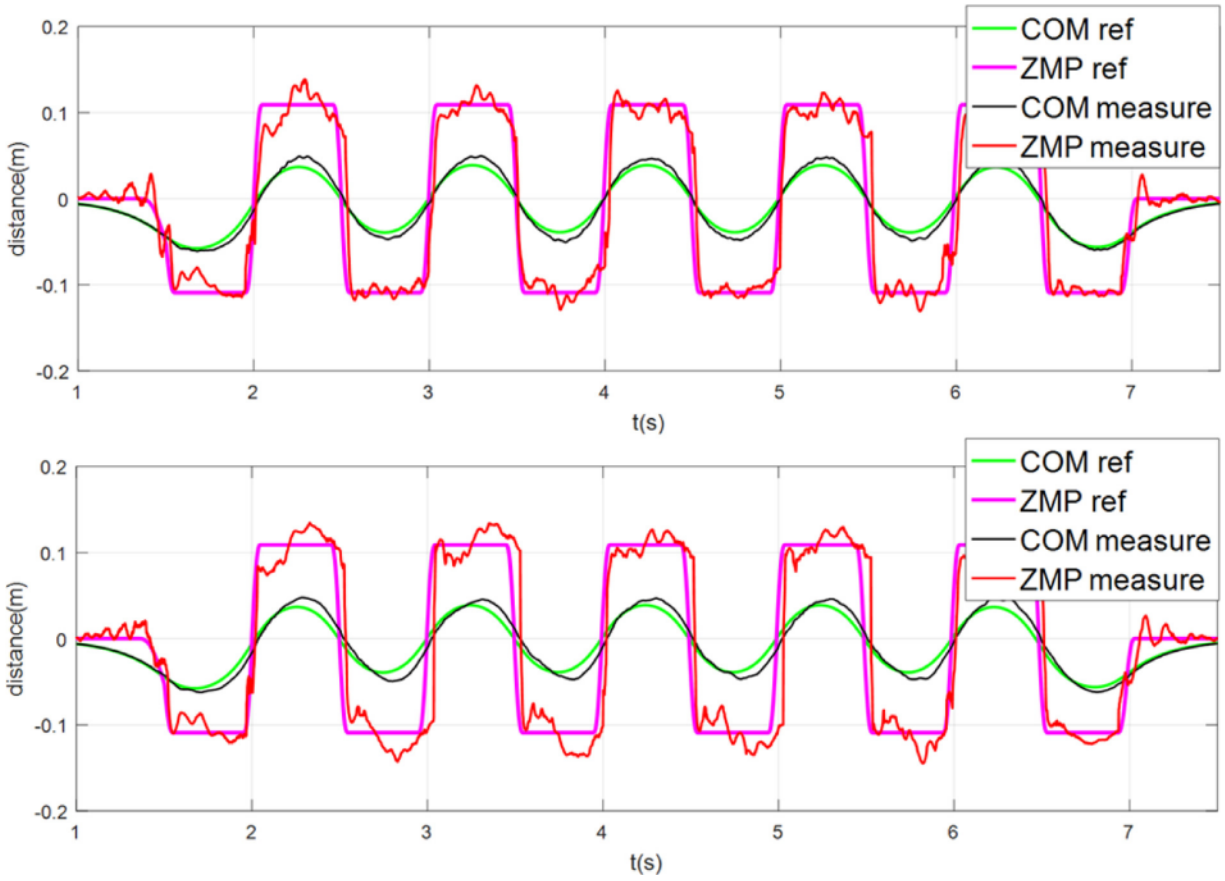


Fig. 19. Plot of reference ZMP and measured ZMP in the y-direction: GAZELLE leg case (top) and conventional leg case (bottom).

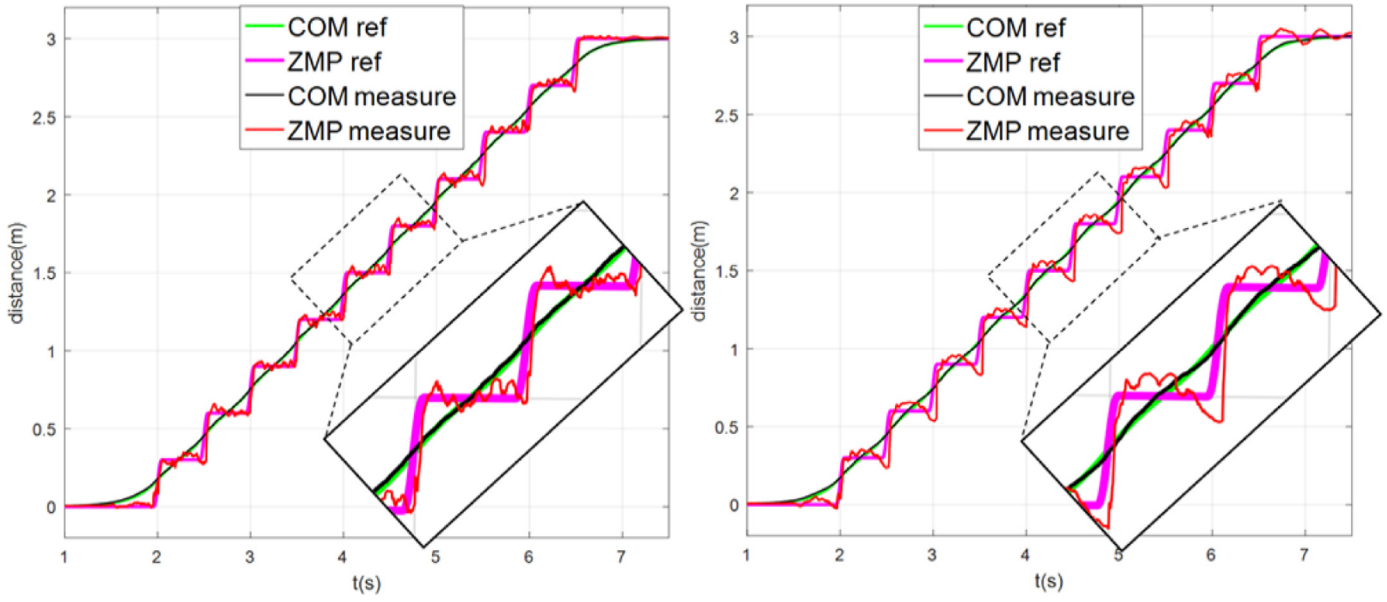


Fig. 20. Plot of reference ZMP and measured ZMP in the *x*-direction: GAZELLE leg case (left) and conventional leg case (right). Simulation with conventional leg model shows more fluctuation of measured ZMP, which originates from the swing leg effect.

4.1. Walking pattern generation

Walking patterns comprise CoM and foot trajectory. The foot trajectory can be easily generated from the position and time schedule of footprints. We used the preview control approach to generate the CoM trajectory [30]. The preview controller has no boundary when designing the reference ZMP trajectory as long as it is known beforehand. Therefore, CoM trajectories can be freely generated for any ZMP trajectory, such as dual-support phase and heel-to-toe walking. For the real-time step adjustment, the CoM trajectory must be continuously and stably

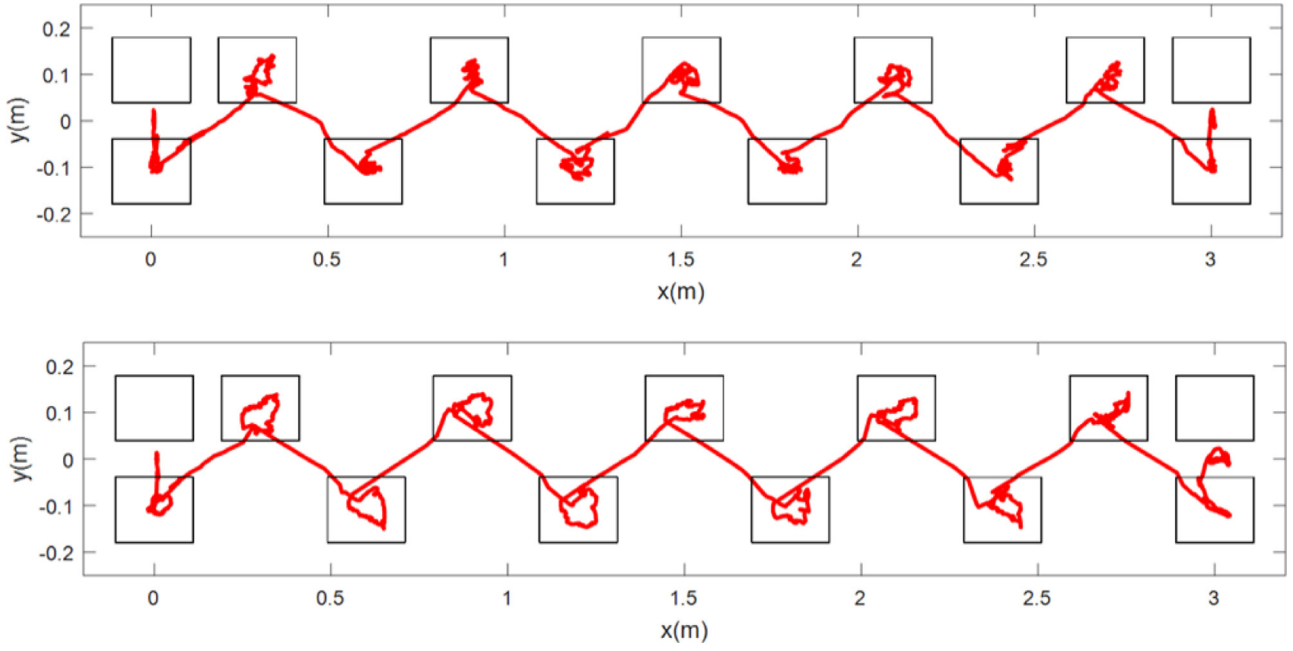


Fig. 21. *x-y* plane plot of measured ZMP: GAZELLE leg case (top) and conventional leg case (bottom).

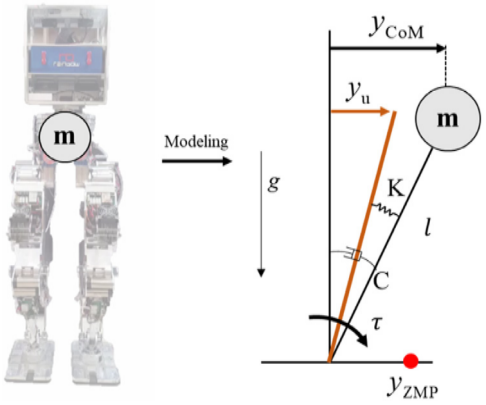


Fig. 22. Compliant LIPM modeling of real robot. In the real situation robot does not behave like ridged body due to joint, structure stiffness and compliance between the ground and the foot.

generated for the sudden change in footprint. Therefore, we utilized a short-cycle preview controller that can adapt a rapid change of footprint [38].

4.2. CoM damping control

If a sudden external force is applied to the standing robot, it will oscillate with a damped sinusoidal form because of the stiffness of each leg joint, including the stiffness of the structure, and the rubber pad attached under the feet (Fig. 23(a)). When the robot is standing with two or one feet, it can be modeled by an inverted pendulum with a spring and a damper [39] (Fig. 22). From the system ID, the GAZELLE has a natural frequency of approximately 2 Hz and a damping ratio of less than 0.1 in the lateral direction. The CoM of the robot easily vibrates even with small impacts because of the small damping ratio; hence, the robot easily becomes unstable during walking. The stiffness should be decreased, and the damping ratio should be increased using a state feedback controller. Accordingly, a mathematical model of the compliant linear inverted pendulum model (LIPM) is needed to construct the controller. We can set up the modeling equation of the compliant LIPM as follows:

$$\begin{aligned} m\ddot{y}_{CoM} &= mgy_{CoM}/l - \tau/l \\ \tau/l &= K(y_{CoM} - y_u) + C\dot{y}_{CoM} \\ y_{ZMP} &= \tau/mg \end{aligned} \tag{27}$$

Eq. (27) is induced from moment equilibrium equation of modeling in Fig. 22. τ/mg is definition of ZMP and K and C are the system's natural stiffness and damping coefficient in y-direction respectively and l is the height of LIPM. We consider herein y_u as the system input which is the CoM position command input. The y_{CoM} is the actual CoM position in y-direction which is assumed that it attached on input stick by spring and damper (Fig. 22). The system outputs are the ZMP, actual CoM position (y_{CoM}), and CoM velocity. The main idea of compliant LIPM is that there is a delay between the CoM command and the actual CoM because of the stiffness and damping of system. The system can be modeled to state space form with state of the CoM position and velocity.

$$\begin{aligned} &\dot{x} = Ax + Bu \quad y = Cx + Du = \begin{bmatrix} y_{ZMP} & y_{CoM} & \dot{y}_{CoM} \end{bmatrix}^T \\ &A = \begin{bmatrix} 0 & 1 \\ g/l - K/m & -C/m \end{bmatrix} B = \begin{bmatrix} 0 \\ K/m \end{bmatrix} x = \begin{bmatrix} y_{CoM} \\ \dot{y}_{CoM} \end{bmatrix} \\ &C = \begin{bmatrix} K \cdot l/mg & C \cdot l/mg \\ 1 & 0 \\ 0 & 1 \end{bmatrix} D = \begin{bmatrix} -K \cdot l/mg \\ 0 \\ 0 \end{bmatrix} u = y_u \end{aligned} \tag{28}$$

The control input u, which is y-directional CoM input command, should be designed from the full-state feedback to specify new stiffness and damping to the original system. Therefore, we need the state estimated from the measurement. The ZMP and the CoM state can be measured by using the F/T sensor and the IMU. We have multiple measurements; hence, we can utilize the Kalman filter. (29) is the formulation of the steady-state Kalman filter used to estimate the CoM state.

$$\begin{aligned} &\dot{\hat{\mathbf{x}}} = \mathrm{A}\hat{\mathbf{x}} + \mathrm{Bu} + L(\mathbf{y}_{\mathrm{measure}} - \mathbf{y}) \\ &\mathbf{y}_{\mathrm{measure}} = \begin{bmatrix} y_{\mathrm{ZMP}}^{\mathrm{measure}} & y_{\mathrm{CoM}}^{\mathrm{measure}} & \dot{y}_{\mathrm{CoM}}^{\mathrm{measure}} \end{bmatrix}^{\mathrm{T}} \\ &\mathrm{u} = -\mathrm{K}\hat{\mathbf{x}} \end{aligned} \tag{29}$$

where L is the Kalman gain obtained from the Riccati equation, and $\mathbf{K}$ is the full-state feedback gain. Fig. 23(b) presents the result of the CoM damping controller.

The most unique feature of the proposed damping controller is that we use IMU information (measured COM position and velocity [40]) and ZMP instead of using ZMP only. The previous HUBO series used a

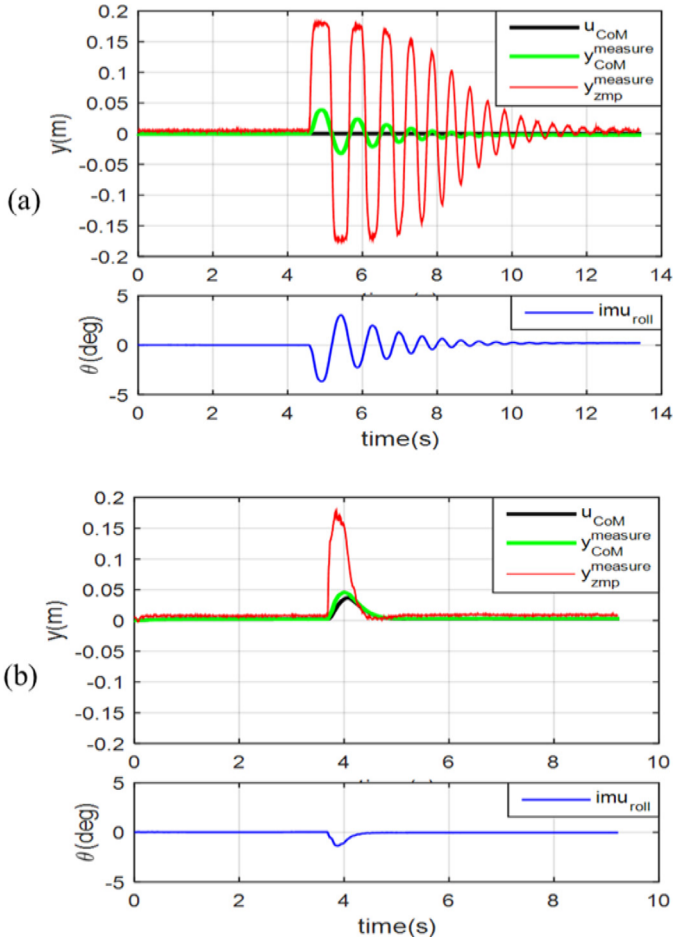


Fig. 23. CoM damping control result. The robot is pushed in the lateral direction while standing. Graphs show the sensor values and the CoM position of the robot. If the robot gets impact, its body vibrates because of high stiffness (a). The CoM damping control raises system damping and becomes robust to impact (b).

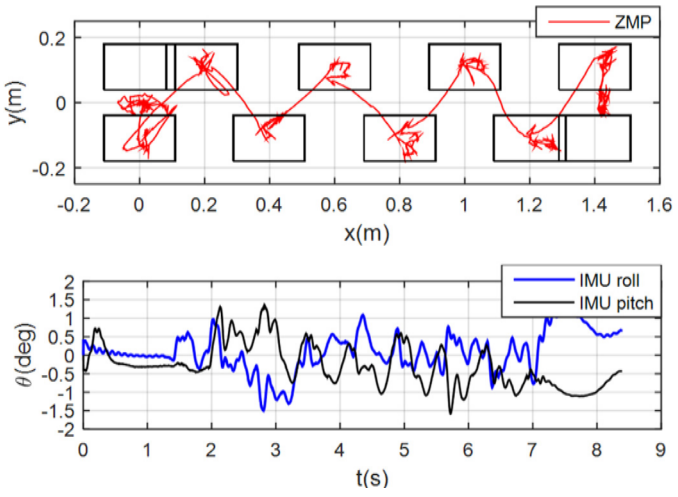


Fig. 24. Experiment data of GAZELLE (uneven terrain walking): *x-y* plane plot of footprints and ZMP (top). IMU angle plot during uneven terrain walking (bottom).

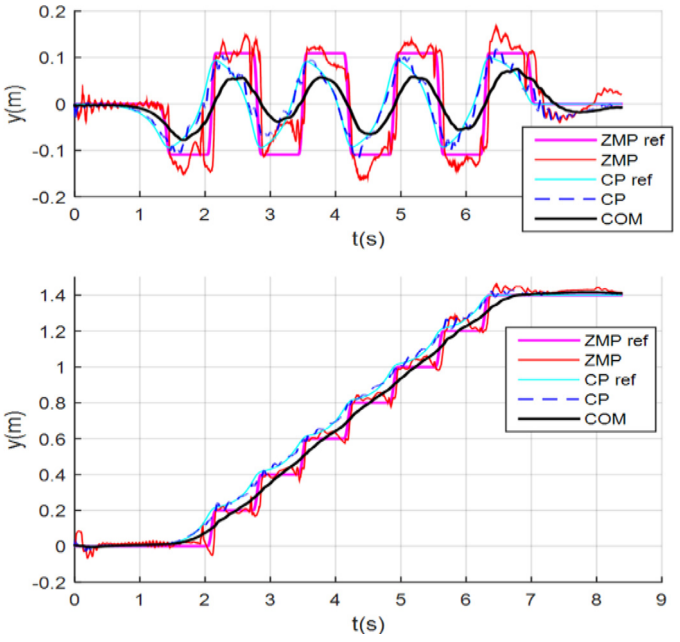


Fig. 25. Time domain plot of uneven terrain walking: *Y*-direction plot (top) and *X*-direction plot (bottom). ZMP ref is reference ZMP generated by preview controller and ZMP is measured ZMP from F/T sensor. CP ref is capture point reference generated by preview controller and CP is measured capture point from IMU. COM is measured center of mass from IMU.

similar form of damping controller, but it used only the measured ZMP from the F/T sensor [41]. Here, we expanded the Luenberger observer form to a Kalman filter that could accept multiple input values. The state observer using only ZMP information is valid when the robot is not tilted. If the robot is tilted (tip-over situation), the ZMP information will not change because the ZMP is saturated at the edge of the foot. Therefore, a state observer cannot recognize the robot's state variation. In such a situation, our proposed form of CoM damping controller can detect the change in the robot's state by using IMU information. Hence, the overall performance of the controller is improved.

4.3. Foot force/torque control

Foot force and torque control are necessary for the robot to walk on uneven ground. To implement this, the reference force and torque of the foot should be obtained from the reference trajectory of the robot and the current state. In [40], we proposed a method for generating the reference torque of a foot using capture point feedback. The error between the reference capture point and the measured capture point led us to obtain the desired ZMP (cZMP). This cZMP was used to generate the ankle torque reference. The reference ankle torque was obtained, and the ankle torque control method was described in Section III.B.

Foot force control was performed in the vertical direction using leg length control. The cZMP from the capture point feedback was used to generate the reference vertical force of both feet [40]. This was then transformed to the desired length of both legs [31]. Fig. 26 shows the total control scheme of GAZELLE.

4.4. Walking experiments

A walking experiment was performed by the GAZELLE platform and the proposed walking algorithm (Figs. 27). In the experiment, GAZELLE walked through the uneven terrain consisting of wood blocks with different heights (0.5 cm–3 cm). The local tilt of the uneven terrain was 5° to 10°. GAZELLE can adapt the height difference and the local tilt using foot force/torque control. CoM damping control absorbed most of the

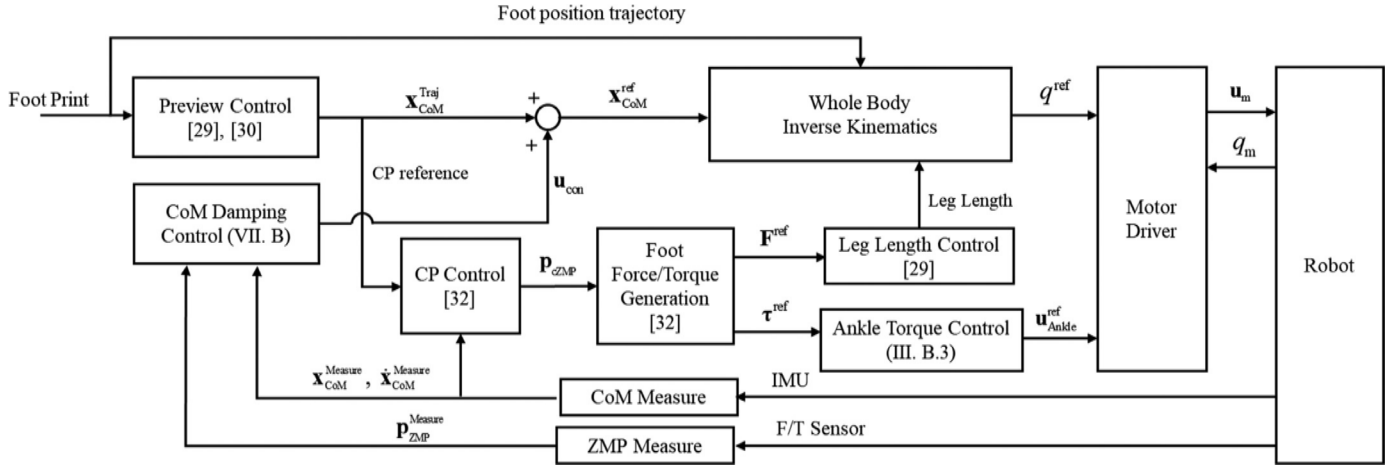


Fig. 26. Overall algorithm for walking control of GAZELLE.

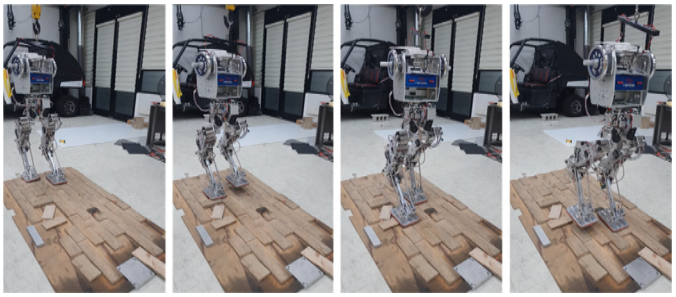


Fig. 27. Snapshot of uneven terrain walking. Uneven terrain is consist with wood block of different height. Range of local tilt angle is 5~10°.

vibrations during walking; hence, the small amount of early landing or late landing can be compensated for. Figs. 24, 25 displays the plots of the walking experiment.

In [42], experiments using the GAZELLE platform are described in detail. [42] introduces the gait stabilization algorithm using ankle, hip and stepping strategy optimization when the robot is disturbed. In this study, we can see the result of overcoming the disturbance by fast stepping when the GAZELLE platform is disturbed.

5. Conclusion

In this study, we designed the lower-body platform, GAZELLE, which was lightweight and capable of a high power output. Lightweight actuators and an efficient structure design reduced its weight. It can also reduce the moment of inertia of the lower body using the link structure. By using the link structure for ankle joint, the stiffness of the joint is lowered and the joint backlash would be occurred. However, in the case of the ankle joint, the joint stiffness and backlash do not affect overall robot performance because the robot uses torque control rather than the position control fop the ankle joint. We verified, through simulations, that reducing the moment of inertia of the leg was effective for walking. It improved the performance of the LIPM-based gait pattern generator and controller, and enabled more rapid foot movement because of the lightness of the legs.

We described herein the kinematics and Jacobian of a 2-RSU parallel ankle actuator in detail. We also suggested a novel air-cooling module design that efficiently dissipated the motor heat. We presented an efficient fin design that could achieve optimal heat dissipation by applying the given conditions.

We validated the overall performance of the platform and the walking algorithm through a walking experiment on uneven terrain. In addition, the hardware design of GAZELLE was verified to be suitable for overcoming external disturbances by rapid movement of the legs, through the experiments shown in [42].

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Hyobin Jeong: Conceptualization, Methodology, Software, Writing - original draft. **KangKyu Lee:** Data curation, Visualization, Validation. **Wooshik Kim:** Writing - original draft, Investigation. **Inho Lee:** Writing - review & editing, Supervision. **Jun-Ho Oh:** Conceptualization, Validation, Funding acquisition.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.mechatronics.2019.102319.

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